



HEXAROOT SYSTEMS

PHASE 1

Lab Setup & Environment Design

A Comprehensive Attack–Defense Simulation Model for DevSecOps Security Hardening

What We Built

01

Private Network

hexaroot-net
192.168.100.0/24

02

Target Containers

DVWA + MySQL +
Ubuntu Internal

03

SIEM Stack

Elasticsearch +
Kibana Dashboard

04

Attacker Machine

Kali Linux VM
VirtualBox

Result: 5 containers running, private lab network operational, SIEM dashboard live at localhost:5601

Prerequisites & Tools

System Requirements		Software to Install	
RAM	8 GB minimum (16 GB recommended)	Docker Desktop	Runs all lab containers
Disk	80 GB free space	VirtualBox	Hosts Kali Linux VM
CPU	64-bit with VT-x / AMD-V enabled	Kali Linux ISO	Attacker machine image
OS	Windows 10/11, Ubuntu or macOS	VS Code / Notepad	Edit docker-compose.yml

Important: Enable Intel VT-x or AMD-V in BIOS before installing VirtualBox

Project Folder & Docker Compose File

Folder Structure

```
C:\hexaroot-lab\  
├── docker-compose.yml  
└── data\  
    ├── mysql\  
    └── wazuh\
```

Windows CMD Commands

```
mkdir C:\hexaroot-lab\data\mysql
```

```
mkdir C:\hexaroot-lab\data\wazuh
```

```
cd C:\hexaroot-lab
```

```
notepad docker-compose.yml
```

```
docker compose up -d
```

```
docker compose ps
```

The docker-compose.yml file is the single blueprint that defines all 5 containers and the network

STEP 3

Docker Network — hexaroot-net

hexaroot-net · 192.168.100.0/24 · Bridge Driver · Gateway: 192.168.100.1

Web App

192.168.100.10

Port: 8080

hexaroot-dvwa

MySQL DB

192.168.100.11

Port: 3306

hexaroot-db

Ubuntu SSH

192.168.100.12

Port: 2222

hexaroot-internal

Elasticsearch

192.168.100.20

Port: 9200

hexaroot-elastic

Kibana SIEM

192.168.100.21

Port: 5601

hexaroot-siem

Network is created AUTOMATICALLY by Docker Compose — no manual docker network create needed

STEP 4

The 5 Containers — Roles & Images

01

hexaroot-dvwa

vulnerables/web-dvwa

ATTACK TARGET

Vulnerable web app — primary attack target

02

hexaroot-db

mysql:5.7

DATA TARGET

MySQL database — stores sensitive employee data

03

hexaroot-internal

ubuntu:22.04

LATERAL TARGET

Internal Ubuntu server with SSH — lateral movement target

04

hexaroot-elastic

elasticsearch:7.17.10

SIEM BACKEND

Log storage engine — receives logs from all containers

05

hexaroot-siem

kibana:7.17.10

SOC DASHBOARD

Web dashboard — visualize logs and detect attacks

STEP 5

DVWA — Vulnerable Web Application

Image: `vulnerables/web-dvwa`

IP: 192.168.100.10 · Host Port: 8080 · Access: <http://localhost:8080>

Vulnerabilities Available for Attack

SQL Injection

File Upload Exploit

Brute Force Login

XSS

Command Injection

CSRF

Setup Steps After Container Starts

- 1 Open <http://localhost:8080> in browser
- 2 Login: admin / password
- 3 Click "Create / Reset Database"
- 4 Login again (auto redirect)
- 5 Go to DVWA Security → set to Low
- 6 All vulnerabilities now accessible

Setting security to LOW is mandatory — without it, vulnerabilities are blocked and attacks won't work

MySQL Database & Internal Ubuntu Server

MySQL Database · hexaroot-db

Image: mysql:5.7

IP: 192.168.100.11

Port: 3306

Root PW: root_secret_2026

DB Name: hexaroot_data

App User: dvwa / dvwa_pass_2026

Stores fake employee PII data (name, Aadhaar, salary) — the "stolen data" for Phase 2

Internal Server · hexaroot-internal

Image: ubuntu:22.04

IP: 192.168.100.12

SSH Port: 2222 (host) → 22 (inside)

User: hexaadmin / admin123

Root PW: toor (weak – intentional)

Secret files in /opt/confidential/secrets.txt

SSH enabled with weak credentials — simulates a misconfigured internal server for lateral movement demo

SIEM Stack — Elasticsearch + Kibana

Original plan used Wazuh all-in-one — replaced with ELK Stack (Elasticsearch + Kibana) after Wazuh deprecated their single Docker image

Elasticsearch • hexaroot-elastic

Image:	elasticsearch:7.17.10
IP:	192.168.100.20
Port:	9200
Mode:	single-node (lab)
Memory:	512 MB capped (ES_JAVA_OPTS)
Auth:	Disabled (xpack.security=false)

Kibana • hexaroot-siem

Image:	kibana:7.17.10
IP:	192.168.100.21
Port:	5601
Access:	http://localhost:5601
Backend:	http://192.168.100.20:9200
Role:	SOC Analyst Dashboard

Kibana takes 2–3 min to initialize. Set `vm.max_map_count=262144` in PowerShell (Admin) before starting

STEP 9

Lab Verification — All Systems Go



`docker compose ps`

All 5 containers show STATUS: Up



`http://localhost:8080`

DVWA login page loads correctly



`http://localhost:9200`

Elasticsearch JSON response shown



`http://localhost:5601`

Kibana dashboard loads



`docker compose logs`

No critical errors in any container

PHASE 1 COMPLETE — Lab is fully operational and ready for Phase 2 Attack Simulation

Phase 1

Complete

What was accomplished:

- Private Docker network (192.168.100.0/24) created
- 5 containers deployed with fixed IPs
- DVWA configured as vulnerable web target
- MySQL seeded with fake sensitive data
- Internal Ubuntu server with SSH + secrets
- ELK Stack SIEM running at localhost:5601
- All containers verified — fully operational

Next: Phase 2

- 1 Reconnaissance with Nmap
- 2 Brute force with Hydra
- 3 SQL Injection with SQLMap
- 4 Reverse Shell via Metasploit
- 5 Lateral Movement to Internal
- 6 Data Exfiltration via Netcat